Literature Review - Smart Academic Communication Platform

About SACP. SACP (Smart Academic Communication Platform) helps a university send the right message to the right students at the right time, using academic records, templates, scheduled campaigns and rules. It has three user roles: **Admin**, **Faculty Member** and **Student**. Each table compares one existing app with SACP's components. **C1 to C14** are core components and **O1 to O4** are optional ones; both lists are at the end. Details are based on 2026 product documentation and update notes.

1. Slack

Team chat app. Channel-based chat for work teams, with a no-code workflow builder and AI add-ons.

Area	What Slack has	What it lacks	What SACP provides and how it fills the gap
Login, roles and 2FA (C1, C13)	Two-factor login, single sign-on and admin controls. Roles such as owner, admin and member.	No Student or Faculty roles. 2FA is for login only, not for sensitive actions.	Admin, Faculty Member and Student roles. 2FA at login and before critical actions like sending a campaign.
Academic data and audience (C2, C4)	Channels people join or are added to.	No course, department or student records. Members are added by hand.	Academic Data Module stores course and department data, so the right students are picked from records.
Templates and campaigns (C3, C4)	Can schedule a single message. Workflow Builder can post on a timetable.	No message templates with placeholders. No campaign tool or database-driven audience.	Reusable templates with placeholders such as name and course. Campaigns are scheduled to an audience chosen from the database.
Automation rules (C5)	Workflow Builder: no-code steps, forms and AI steps (limited on cheaper plans).	Triggers are Slack events, not academic ones. No monitoring screen for campaigns.	Rules made by users react to academic events, with a screen to manage and watch each workflow.
Email, alerts and analytics (C6, C9)	Push and email alerts about chat activity. Basic usage stats.	Not an email outreach tool. No delivery or open tracking for notices.	Email delivery with tracking, campaign alerts and event reminders, plus a dashboard for admins and faculty.
1-to-1 chat and files (C7, C10)	Direct messages, threads, huddles and file sharing. Very strong.	General chat, not set up for student queries and feedback.	Matches this, but keeps 1-to-1 chat and attachments focused on student-faculty queries.
Smart sorting, bot and routing (C8, C11, C14)	Slackbot and Slack AI answer questions, search and summarise.	No spam or priority sorting of student queries. No routing to the right staff member.	ML filters spam and ranks urgent queries. FAQ chatbot answers common questions. AI Query Evaluator sends each query to the right staff.
Polls and surveys (C12)	Polls through add-on apps and workflow forms.	No built-in anonymous survey.	Quick polls and anonymous surveys built in.
Optional ideas (O1 to O4)	Slack AI drafts and summarises (paid). Emoji reactions and schedule send. Calendar apps are available.	No academic deadline export. No send-time choice based on when students are free.	Optional: AI drafting and summaries, calendar export, reactions and send-time optimization, all tied to academic data.

Summary: Slack vs SACP. Slack is excellent for team chat and has useful AI and automation, but it knows nothing about courses, students or grades. SACP covers the academic side: records-based audiences, templates, campaigns, tracked email and query routing.

2. WhatsApp

Phone messaging app. Used by almost every student. Has groups, broadcast lists, Channels and a separate Business platform.

Area	What WhatsApp has	What it lacks	What SACP provides and how it fills the gap
Login, roles and 2FA (C1, C13)	Phone-number login, optional two-step verification, end-to-end encryption. Group admins.	No institution roles. The university cannot control who is Admin, Faculty or Student.	Role-based login for Admin, Faculty and Student. 2FA at login and for critical actions.
Academic data and audience (C2, C4)	Phone contacts. Groups hold up to 1,024 people. Broadcast lists hold 256 and only reach people who saved the sender.	No academic records. Faculty must save every student's number.	Audience comes from stored course and department records, not anyone's phonebook.
Templates and campaigns (C3, C4)	Broadcast lists. Business platform has approved message templates and bulk sending.	Business platform needs a developer, opt-in and template approval. No scheduling in the normal app.	Staff create templates and schedule campaigns from one screen, with no approval steps outside the system.
Automation rules (C5)	Auto-reply and quick replies in the Business app.	No rules based on academic events.	Automation Rule Engine triggers campaigns from user-defined rules.
Email, alerts and analytics (C6, C9)	Delivery ticks. Business platform shows delivery and read stats.	No email. Normal accounts get no message stats.	Email delivery with tracking, alerts and reminders, and a dashboard of results.
1-to-1 chat and files (C7, C10)	Fast 1-to-1 chat, calls and files up to 2 GB.	Mixes personal and study life. Faculty must share personal numbers. No institutional record.	1-to-1 chat inside the institution, built for queries and feedback. Attachments are shared between accounts.
Smart sorting, bot and routing (C8, C11, C14)	Business app quick replies. Chatbots only through the paid platform.	No priority sorting or routing of student questions.	ML sorting, FAQ chatbot and AI routing to the right staff.
Polls and surveys (C12)	Polls in chats and groups.	No survey tool.	Quick polls and anonymous surveys.
Optional ideas (O1 to O4)	Emoji reactions.	No calendar export, AI drafting for announcements or send-time choice.	Optional: AI drafting, calendar export, reactions and send-time optimization.

Summary: WhatsApp vs SACP. WhatsApp reaches everyone but has no link to academic data, no staff-friendly campaign tools, and no way to sort or route queries. SACP adds a controlled, institution-owned space for the same conversations.

3. Microsoft Teams

Chat, meetings and classes. Part of Microsoft 365. Education version adds class teams, Assignments, Insights and Copilot.

Area	What Microsoft Teams has	What it lacks	What SACP provides and how it fills the gap
Login, roles and 2FA (C1, C13)	Microsoft login with MFA. Owner, member and guest roles. Teacher and student roles in Education.	MFA is set for the whole organisation, not for single sensitive actions. Tied to Microsoft accounts.	Own role set (Admin, Faculty, Student). 2FA also required before critical actions like sending campaigns.
Academic data and audience (C2, C4)	School Data Sync builds class teams from student records. Strongest of all apps here.	Needs Microsoft 365 Education setup. Audience mostly limited to class teams.	Academic Data Module holds courses and departments, and any campaign can pick its audience from it.
Templates and campaigns (C3, C4)	Schedule send for chats and channel posts. Announcement posts.	No templates with placeholders. No campaign tool.	Templates with placeholders. Campaigns scheduled to a database-selected audience.
Automation rules (C5)	Power Automate and Workflows for custom flows.	General office tool. Needs setup and is not made for academic campaigns.	Simple rule screen made for academic campaigns, with monitoring.
Email, alerts and analytics (C6, C9)	Chat and email notifications. Education Insights shows class engagement.	No delivery or open tracking for notices.	Email tracking, alerts, reminders and a communication dashboard.
1-to-1 chat and files (C7, C10)	Chat, calls and shared files. Very strong.	General chat. Student queries mix with everything else.	Matches this, but focused on student-faculty query and feedback.
Smart sorting, bot and routing (C8, C11, C14)	Copilot summarises chats and drafts replies.	No spam or priority sorting of queries. No routing to staff.	ML sorting, FAQ chatbot and AI query routing.
Polls and surveys (C12)	Microsoft Forms and Polly for polls and quizzes (separate apps).	Needs extra apps.	Polls and anonymous surveys built into the platform.
Optional ideas (O1 to O4)	Copilot drafting and summaries. Outlook calendar. Reactions. Schedule-send suggestions for the recipient's work hours.	Needs Microsoft licences. Not designed around academic deadlines.	Optional: the same four ideas, bundled with academic data and no Microsoft dependency.

Summary: Microsoft Teams vs SACP. Teams for Education is the closest match here, with roster sync, scheduling and AI help. It is still a general collaboration suite, so SACP stays ahead on campaign tools, rule-based automation, tracking and query routing.

4. Google Classroom

Class management tool. Teachers run classes, assignments, grades and announcements. Gemini AI is now built in.

Area	What Google Classroom has	What it lacks	What SACP provides and how it fills the gap
Login, roles and 2FA (C1, C13)	Google login with 2-step verification. Teacher, student and guardian roles.	Roles belong to a class. No institution-wide Admin role for communication. 2FA not tied to posting.	Admin, Faculty and Student roles across the institution. 2FA before critical actions.
Academic data and audience (C2, C4)	Classes with student lists, assignments and grades.	Audience is the class list. No department-wide or database-driven selection.	Audience is chosen from academic records across courses and departments.
Templates and campaigns (C3, C4)	Announcements and assignments can be scheduled. Old posts can be reused.	No placeholders. No campaign tool.	Templates with placeholders. Scheduled campaigns.
Automation rules (C5)	Automatic email notifications and due-date reminders.	No user-defined rules.	Rule engine lets users define their own triggers and actions.
Email, alerts and analytics (C6, C9)	Email alerts, guardian summaries and an Analytics tab for missing work and at-risk students.	Analytics covers coursework, not who opened a notice.	Campaign analytics: delivery, opens and results.
1-to-1 chat and files (C7, C10)	Class and private comments, Drive attachments.	No real-time 1-to-1 chat.	Real-time 1-to-1 chat with file sharing.
Smart sorting, bot and routing (C8, C11, C14)	Gemini gives insights and answers from teacher materials.	No spam or priority sorting of queries, no routing, no FAQ bot for institution questions.	ML sorting, FAQ chatbot and routing to the right staff.
Polls and surveys (C12)	Question posts and Google Forms.	Surveys need Forms.	Built-in polls and anonymous surveys.
Optional ideas (O1 to O4)	Gemini drafts posts. Google Calendar links due dates.	No reactions on announcements. No send-time choice.	Optional: AI drafting, calendar export, reactions and send-time optimization.

Summary: Google Classroom vs SACP. Classroom is great for running a class, and Gemini has narrowed the gap on AI help. It remains class-bound and has no campaigns, rules, tracked email or query routing, which SACP adds.

5. Discord

Community chat app. Servers with text, voice and forum channels. Built for communities, not for schools.

Area	What Discord has	What it lacks	What SACP provides and how it fills the gap
Login, roles and 2FA (C1, C13)	Login with optional 2FA (can be required for moderators). Server roles and permissions.	Roles are made by hand and do not match Admin, Faculty and Student. Not linked to university identity.	Institution roles and 2FA before critical actions.
Academic data and audience (C2, C4)	Servers and roles.	No academic records. Membership by invite.	Audience comes from stored academic data.
Templates and campaigns (C3, C4)	Announcement channels. Scheduled events with RSVP.	No templates or campaigns. Scheduled messages need bots.	Templates and scheduled campaigns built in.
Automation rules (C5)	AutoMod rules and custom bots.	Bots need coding. No academic rules.	User-defined rules with a simple management screen.
Email, alerts and analytics (C6, C9)	App notifications. Server Insights shows growth and activity.	No email. No delivery tracking of notices.	Tracked email, alerts, reminders and dashboard.
1-to-1 chat and files (C7, C10)	Direct messages, threads, forum channels and voice. Free accounts have a small file limit.	Not organised for query and feedback between students and faculty.	1-to-1 chat and file sharing focused on academic queries.
Smart sorting, bot and routing (C8, C11, C14)	AutoMod filters spam. AI conversation summaries.	No query priority or routing. FAQ bot needs a custom bot.	ML sorting, FAQ chatbot and routing.
Polls and surveys (C12)	Polls with up to 10 options.	No anonymous survey.	Polls and anonymous surveys.
Optional ideas (O1 to O4)	Emoji reactions. Scheduled events.	No AI drafting for faculty, calendar export or send-time choice.	Optional: AI drafting, calendar export, reactions and send-time optimization.

Summary: Discord vs SACP. Discord is lively and flexible for open communities, but it has no academic data, no email tools and no query handling. SACP is controlled, records-based and aimed at official communication.

6. Moodle

Open-source learning platform. Full learning platform with courses, forums, quizzes, grades and notifications.

Area	What Moodle has	What it lacks	What SACP provides and how it fills the gap
Login, roles and 2FA (C1, C13)	Login with MFA (email, SMS, security key or authenticator app). Detailed roles and permissions.	Roles take effort to set up. MFA applies to login, not to sending a campaign.	Three simple roles. 2FA also protects critical admin and faculty actions.
Academic data and audience (C2, C4)	Courses, groups, cohorts and grades.	Messages follow course context. Hard to target by department or rule.	Audience picked from academic records across courses and departments.
Templates and campaigns (C3, C4)	Site announcements and course news forums.	Default message wording. No placeholder templates or campaign scheduling.	Templates with placeholders and scheduled campaigns.
Automation rules (C5)	Event monitoring rules and automatic notices for deadlines, grades and forum posts.	Limited to Moodle events and a basic setup screen.	Rules with a friendly screen to manage and monitor workflows.
Email, alerts and analytics (C6, C9)	Email, app push and SMS notifications (since version 5.0). Report Builder.	No open or click tracking for notices.	Tracked email delivery, reminders and a communication dashboard.
1-to-1 chat and files (C7, C10)	Built-in messaging, forums and file uploads.	Plain messaging without priority or query focus.	Focused 1-to-1 chat for queries and feedback with attachments.
Smart sorting, bot and routing (C8, C11, C14)	AI tools for drafting and summaries, with course-level controls. Chatbots only by plugin.	No spam filter or priority sorting. No routing of queries.	ML sorting, built-in FAQ chatbot and AI query routing.
Polls and surveys (C12)	Choice (poll) and Feedback (survey) activities.	Course setup needed. Not a quick poll inside chat.	Quick polls and anonymous surveys in the messaging flow.
Optional ideas (O1 to O4)	AI drafting. Calendar export to personal calendars.	No reactions or send-time choice.	Optional: AI drafting, calendar export, reactions and send-time optimization.

Summary: Moodle vs SACP. Moodle is the strongest at teaching and has real notification tools, including SMS. It is not a communication platform: campaigns, templates, tracking and query routing are missing, and SACP can work beside it.

7. Gmail

Email service. Google's email with labels, filters, scheduling and Gemini AI. Mail merge needs a paid Workspace plan.

Area	What Gmail has	What it lacks	What SACP provides and how it fills the gap
Login, roles and 2FA (C1, C13)	Google login with 2-step verification. Workspace admin roles.	No academic roles. 2-step verification covers login only.	Admin, Faculty and Student roles. 2FA also before critical actions.
Academic data and audience (C2, C4)	Contacts, groups and labels.	No academic records. Class lists kept by hand.	Audience from stored academic data.
Templates and campaigns (C3, C4)	Templates, Schedule send and mail merge (up to 1,500 a day on Workspace).	Schedule send is one email at a time with no repeats. Mail merge cannot be scheduled. Daily send limits.	Templates with placeholders. Scheduled campaigns to database-selected audiences.
Automation rules (C5)	Filters work on incoming mail.	No rules that send messages when an academic event happens.	Rule engine sends campaigns from academic events.
Email, alerts and analytics (C6, C9)	Email delivery. Read receipts only on some Workspace accounts.	No open or click dashboard. No in-app alerts.	Email tracking, campaign alerts, reminders and dashboard.
1-to-1 chat and files (C7, C10)	Email threads with attachments up to 25 MB.	No real-time chat. Queries mix with other mail.	Real-time 1-to-1 chat and file sharing.
Smart sorting, bot and routing (C8, C11, C14)	Strong spam filter. AI Overview summaries and suggested replies.	No priority for student queries and no routing to staff. No FAQ bot.	ML tuned for academic queries, FAQ chatbot and routing.
Polls and surveys (C12)	None built in.	Needs Google Forms.	Built-in polls and anonymous surveys.
Optional ideas (O1 to O4)	Gemini Help me write and summaries. Google Calendar links. Emoji reactions.	No automatic send-time choice.	Optional: AI drafting, calendar export, reactions and send-time optimization in one place.

Summary: Gmail vs SACP. Gmail is easy and familiar, with good AI writing help, but it handles one email at a time and knows nothing about students. SACP gives institutions campaigns, rules, tracking and chat on top of academic data.

SACP Components

Core components	What it does
1. Auth	Secure user login and role-based access control.
2. Academic Data Module	Stores academic records and course/department data to streamline workflows and target the right audience.
3. Template Management	Reusable message templates with dynamic placeholders for personalized bulk communication.
4. Campaign Management	Campaign creation, scheduling and database-driven target audience selection.
5. Automation Rule Engine (ARE)	Automates campaigns using user-defined rules, with a user-friendly interface to manage and monitor workflows.
6. Messaging and Notification System	Email delivery and tracking, plus campaign alerts and event reminders.
7. Real-time (1-to-1) Communication	One-to-one chat for query and feedback communication between students and faculty.
8. Message Classification and Prioritization	Machine learning to filter spam and prioritize critical academic messages like student queries.
9. Analytics and Reporting	Dashboard for administrators and faculty to track and visualize campaign communication metrics.
10. File Sharing Module	Message attachments and document sharing among users.
11. ChatBot	Two-way FAQ chatbot that answers common student questions and frees up faculty.
12. Poll / Surveys	Integrated quick polls and anonymous surveys to gauge class comprehension quickly.
13. 2FA	2FA for critical admin and faculty actions to prevent unauthorized broadcasts and campaigns. Also applies at login to stop hackers.
14. AI-Based Query Evaluator	Evaluates student queries and automatically routes them to the appropriate staff or faculty member.

Optional components (not yet finalized)	What it does
O1. AI-Assisted Message Composition and Summarization	AI assistant to help faculty draft replies and summarize complex announcements into clear bullet points.
O2. Calendar Integration	Integrate with Google Calendar and Outlook to export event reminders and help students track deadlines.
O3. Reactions	Emoji reactions on announcements and replies, a low-friction way to acknowledge messages and gauge engagement.
O4. Send-Time Optimization	Schedule messages when recipients are available (early evening instead of 3 AM) to maximize engagement.